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Inventor(s)	Xu; Jun et al.

Foam proportional mixing device

Abstract

A foam proportional mixing device includes a Roots pump and a gear pump. The Roots pump comprises a Roots pump inlet, a pair of rotors and a fluid guide member. The fluid guide member is arranged in the Roots pump inlet, and is configured to guide fluid to flow to the pair of rotors and provide driving forces to the pair of rotors for rotating same in opposite directions from each other. The gear pump includes a pair of gears. The foam proportional mixing device includes a rotor shaft of one rotor in the pair of rotors of the Roots pump is connected to a gear shaft of one gear in the pair of gears of the gear pump, such that the one rotor drives the one gear to rotate.

Inventors:	**Xu; Jun (Shanghai, CN), Tu; Shijie (Shanghai, CN), Han; Dong (Shanghai, CN), Ju; Zhengxi (Shanghai, CN), Ran; Jinyu (Shanghai, CN), Wang; Chengyu (Shanghai, CN)**
Applicant:	**Tyco Fire Products LP** (Cranston, RI)
Family ID:	**1000008822050**
Assignee:	**Tyco Fire Products LP (Cranston, RI)**
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Primary Examiner: Gorman; Darren W

Attorney, Agent or Firm: Foley & Lardner LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is the U.S. National Stage Application of International Application No. PCT/CN2021/130160, filed Nov. 11, 2021, which claims the benefit of and priority to Chinese Patent Application No. 202011260021.9, filed Nov. 12, 2020, and Chinese Patent Application No. 202022614332.2, filed Nov. 12, 2020, the entire disclosures of which are incorporated by reference herein.

TECHNICAL FIELD

(2) The present disclosure relates to the technical field of fire safety, in particular to a foam proportional mixing device.

BACKGROUND ART

(3) A foam fire extinguishing system is an important facility to ensure fire safety, which is widely used in tunnels, warehouses, oil depots, and other buildings, and has a targeted effect on the prevention of class A, class B, and class F fires. A foam proportional mixing device is a core component of the foam fire extinguishing system. The existing pressure-type foam proportional mixing device takes the Venturi tube as a core component. When a water pump works normally, a proportional mixer sucks a foam concentrate into a fire-fighting pipe by means of the Venturi tube to mix with fire-fighting water. The foam concentrate and the fire-fighting water are sprayed out by means of a spray gun for fire-fighting operations after being mixed.

SUMMARY OF THE INVENTION

(4) The present disclosure provides a foam proportional mixing device, which uses a Roots pump as a core component, and can achieve the function of proportional mixing of foam and water by means of a simple and compact structure. Moreover, the device is convenient to use and can comply with fire-fighting work on various occasions.

(5) In one aspect, the present disclosure provides a foam proportional mixing device, the foam proportional mixing device comprising a Roots pump and a gear pump. The Roots pump comprises a Roots pump housing, a Roots pump inlet and a Roots pump outlet, a pair of rotors, and a fluid guide member. A Roots pump cavity is formed in the Roots pump housing. The Roots pump inlet and the Roots pump outlet are respectively arranged on two opposite sides of the Roots pump housing, and the Roots pump inlet and the Roots pump outlet are respectively in communication with the Roots pump cavity. The pair of rotors are located in the Roots pump cavity. The fluid guide member is arranged in the Roots pump inlet, and is configured to guide a fluid to flow to the pair of rotors and provide driving forces to the pair of rotors for rotating same in opposite directions from each other. The gear pump comprises a gear pump housing and a pair of gears. A gear pump cavity is formed in the gear pump housing, a gear pump inlet and a gear pump outlet are respectively provided on two opposite sides of the gear pump housing, and the gear pump outlet is in fluid communication with the Roots pump cavity. The pair of gears are located in the gear pump cavity. Here, the foam proportional mixing device is configured as follows: a rotor shaft of one rotor in the pair of rotors of the Roots pump is connected to a gear shaft of one gear in the pair of gears of the gear pump, so as to drive the one gear by means of the one rotor to rotate.

(6) According to the foam proportional mixing device described above, the Roots pump further comprises a Roots pump inlet pipe connected to an outer side of the Roots pump housing, and the Roots pump inlet is partially formed in the Roots pump inlet pipe.

(7) According to the foam proportional mixing device described above, the Roots pump further comprises a foam receiving port, wherein the foam receiving port is arranged on the Roots pump inlet pipe, and the foam receiving port is in communication with the gear pump outlet, so as to fluidly communicate the gear pump outlet with the Roots pump cavity.

(8) According to the foam proportional mixing device described above, the foam proportional mixing device further comprises a coupling, wherein the coupling is connected between the rotor shaft of the one rotor and the gear shaft of the one gear, and the coupling, the rotor shaft, and the gear shaft are arranged coaxially, so that the one rotor can drive the one gear to rotate by means of the coupling.
(9) According to the foam proportional mixing device described above, the Roots pump housing has a height direction, the rotor shafts of the pair of Roots pump rotors both extend along the height direction, and the fluid guide member extends along the height direction of the Roots pump housing.
(10) As in the aforementioned foam proportional mixing device, the fluid guide member comprises a pair of fluid guide surfaces, wherein the pair of fluid guide surfaces are configured as follows: when a fluid flows from the Roots pump inlet to the Roots pump cavity, the pair of fluid guide surfaces guide the fluid to form two sub-flows, and the two sub-flows flow away from each other to respectively drive the pair of rotors to rotate in opposite directions from each other.
(11) As in the aforementioned foam proportional mixing device, the fluid guide member is substantially in a triangular prism shape, and the guide member of the triangular prism shape comprises a top surface, a bottom surface, a flow dividing edge extending between the top surface and the bottom surface, and a pair of side surfaces connected to two opposite sides of the flow dividing edge, the pair of side surfaces forming the pair of fluid guide surfaces; wherein the top surface and the bottom surface are respectively connected to an inner wall of the Roots pump inlet, and the flow dividing edge is arranged away from the Roots pump cavity.
(12) According to the foam proportional mixing device described above, the guide member of the triangular prism shape further comprises a side surface opposite to the flow dividing edge, and the side surface opposite to the flow dividing edge is flush with an inner wall of the Roots pump housing at the position of the Roots pump inlet; and the area of the side surface opposite to the flow dividing edge is A, the opening area of the Roots pump inlet at the position of the inner wall of the Roots pump housing is S, and the area A of the side surface and the opening area S satisfy: $\frac{1}{4} \leq A:S \leq \frac{3}{4}$.
(13) According to the foam proportional mixing device described above, the cross-section of the guide member of the triangular prism shape is an isosceles triangle, the pair of fluid guide surfaces correspond to two legs of the isosceles triangle, the length of the base of the isosceles triangle is b, the height of the isosceles triangle is h, and the ratio h:b between the base b and the height h satisfies: $\frac{1}{3} \leq h:b \leq \frac{1}{2}$.
(14) According to the foam proportional mixing device described above, the gear pump inlet is configured to receive a foam concentrate, and the foam pump is configured to suck in the foam concentrate from the gear pump inlet and discharge the foam concentrate from the gear pump outlet; and the Roots pump inlet is configured to receive a pressure fluid and the foam concentrate from the gear pump outlet, and the Roots pump is configured to mix the pressure fluid and the foam concentrate flowing in from the Roots pump inlet pipe in the Roots pump cavity and discharge same from the Roots pump outlet.
(15) In the present disclosure, the Roots pump is applied to the foam proportional mixing device, and the foam proportional mixing device is enabled to have a compact structure by utilizing the small size of Roots pump, so as to facilitate the application of the foam proportional mixing device to various fire-fighting occasions. At the same time, in the present disclosure, a fluid guide member is added to the Roots pump, and the Roots pump can realize the normal rotation of the rotor only under the action of the kinetic energy of the pressure fluid without the need for additional power by utilizing the flow guiding effect of the fluid guide member.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** shows the structure of a foam proportional mixing device of embodiments of the present disclosure.

(2) FIG. **2**A and FIG. **2**B respectively show the internal structure of the foam proportional mixing device in FIG. **1** at different angles;

(3) FIG. **3** is a transverse sectional view of the gear pump in FIG. **1** at the positions of a gear pump inlet and a gear pump outlet;

(4) FIG. **4** is a transverse sectional view of the Roots pump in FIG. **1** at the positions of a Roots pump inlet and a Roots pump outlet;

(5) FIG. **5**A and FIG. **5**B respectively show the internal structure of a lower housing of the Roots pump in FIG. **1** at different angles;

(6) FIG. **6**A to FIG. **6**E respectively show the working states of a pair of rotors in FIG. **4** in one operation cycle; and

(7) FIG. **7**A and FIG. **7**B are respectively longitudinal sectional views of the foam proportional mixing device in FIG. **1** at different angles.

DETAILED DESCRIPTION OF EMBODIMENTS

(8) Various specific embodiments of the present disclosure will be described below with reference to the accompanying drawings, which form a part of the Specification. It should be understood that although directional terms, such as “front,” “rear,” “upper,” “lower,” “left,” “right,” etc., are used in the present disclosure to describe various exemplary structural parts and elements of the present disclosure, these terms used herein are for illustration only and are determined based on the example orientations shown in the drawings. Since the embodiments disclosed in the present disclosure may be arranged in different orientations, these directional terms are for illustration only and should not be regarded as limitations.

(9) FIG. **1** shows the structure of a foam proportional mixing device **100** of embodiments of the present disclosure. As shown in FIG. **1**, the foam proportional mixing device **100** comprises a Roots pump **101** and a gear pump **102**. Positioning is carried out according to the X-axis, Y-axis, and Z-axis directions shown in FIG. **1**, and the gear pump **102** is located above the Roots pump **101** along the Z-axis direction. The gear pump **102** can draw a foam concentrate from an external foam concentrate tank (not shown in the figure), and can deliver the drawn foam concentrate to the Roots pump. The Roots pump **101** can simultaneously receive a fire-fighting water from outside and the foam concentrate from the gear pump **102**, and mix the fire-fighting water and the foam concentrate to form a foam solution.

(10) The gear pump **102** comprises a gear pump housing **113** and a pair of gears **207** (see FIG. **2**A), wherein the pair of gears **207** are accommodated within the gear pump housing **113**. The gear pump housing **113** comprises a gear pump upper cover **114** and a gear pump lower housing **115**. A gear pump inlet **124** and a gear pump outlet **205** are respectively provided on two opposite sides of the gear pump housing **113** in the X-axis direction (see FIG. **2**B). In the embodiments of the present disclosure, both the gear pump inlet **124** and the gear pump outlet **205** are located on the gear pump lower housing **115**. Here, the gear pump inlet **124** is used to connect with the external foam concentrate tank to receive the foam concentrate from the foam concentrate tank, and the gear pump outlet **205** is used to discharge the foam concentrate.

(11) The Roots pump **101** comprises a synchronous gear outer casing **150**, a pair of synchronous gears **201** (see FIG. **2**A and FIG. **2**B), a Roots pump housing **116**, a pair of rotors **203** (see FIG. **2**A and FIG. **2**B), a Roots pump inlet pipe **146**, and a Roots pump outlet pipe **147**, wherein the pair of rotors **203** are accommodated in the Roots pump housing **116**. As shown in FIG. **1**, the synchronous gear outer casing **150** is arranged between the gear pump housing **113** and the Roots pump housing **116** for accommodating the pair of synchronous gears **201**. The synchronous gear outer casing **150** comprises a synchronous gear upper casing **151** and a synchronous gear lower casing **152**, wherein

the synchronous gear upper casing **151** is in contact with the gear pump lower housing **115**. The synchronous gear upper casing **151** and the synchronous gear lower casing **152** jointly define an accommodating space of the pair of synchronous gears **201**.

(12) The Roots pump housing **116** has a height direction. As shown in FIG. **1**, the height direction of the Roots pump housing **116** is consistent with the Z-axis direction. The Roots pump housing **116** comprises a Roots pump upper cover **111** and a Roots pump lower housing **112**, wherein the Roots pump upper cover **111** is in contact with the synchronous gear lower casing **152**. The Roots pump upper cover **111** and the Roots pump lower housing **112** jointly define an accommodating space of the pair of synchronous gears **201**. The Roots pump inlet pipe **146** and the Roots pump outlet pipe **147** are respectively arranged on two opposite sides of the Roots pump housing **116**. In the embodiments of the present disclosure, the Roots pump inlet pipe **146** and the Roots pump outlet pipe **147** are both located on the Roots pump lower housing **112**. Both the Roots pump inlet pipe **146** and the Roots pump outlet pipe **147** are round pipes, and extend in the X-axis direction. The Roots pump inlet pipe **146** is located on the left side of the Roots pump housing **116**, and a Roots pump inlet **103** is formed in a pipe opening of the Roots pump inlet pipe **146** for receiving a pressure fluid to drive the pair of rotors **203** to rotate. In the embodiments of the present disclosure, the pressure fluid is a fire-fighting water. The Roots pump inlet pipe **146** can be connected to an external fire-fighting water supply end by means of an external pipe (not shown in the figure), so that the fire-fighting water from the fire-fighting water supply end can enter the Roots pump inlet **103** by means of the pipe opening of the Roots pump inlet pipe **146**. A foam receiving port **117** is provided on the Roots pump inlet pipe **146**. The foam receiving port **117** is located on an upper surface of the Roots pump inlet pipe **146** and penetrates through a pipe wall of the Roots pump inlet pipe **146**, so that the foam receiving port **117** is in communication with the Roots pump inlet **103**. The foam receiving port **117** has a substantially circular cross-section and can be in communication with the gear pump outlet **205** by means of an external guide pipe (not shown in the figure), so that the foam receiving port **117** can receive the foam concentrate discharged from the gear pump outlet **205**. A Roots pump outlet **104** is formed in a pipe opening of the Roots pump outlet pipe **147**, and the Roots pump outlet **104** is used to discharge a foam mixed water mixed with the foam concentrate and the fire-fighting water. Since the Roots pump inlet pipe **146** and the Roots pump outlet pipe **147** are respectively located on two opposite sides of the Roots pump housing **116**, the Roots pump inlet **103** and the Roots pump outlet **104** are also respectively located on the two opposite sides of the Roots pump housing **116**.

(13) FIG. **2**A and FIG. **2**B respectively show the internal structure of the foam proportional mixing device **100** in FIG. **1** at different angles. In order to illustrate the internal structure of the Roots pump **101** and the gear pump **102** conveniently, the gear pump upper cover **114**, the synchronous gear outer casing **150**, and the Roots pump upper cover **111** are removed from the foam proportional mixing device **100** in FIG. **2**A and FIG. **2**B. As shown in FIGS. **2**A and **2**B, a gear pump cavity **206** is formed in the gear pump housing **113**, and a pair of gears **207** are intermeshed in the gear pump cavity **206**. The pair of gears **207** have the same size and shape, wherein a gear shaft **217** is arranged at a central position of each gear **207**. Two gear shafts **217** respectively extend along the Z-axis direction, and each gear **207** can rotate around its corresponding one of the gear shafts **217**. Since two gears **207** are meshed with each other, when one of the gears **207** actively rotates, the other gear **207** can be driven to rotate accordingly. The volume of the gear pump cavity **206** is matched with the size of the pair of gears **207**, so that when the pair of gears **207** are meshingly rotating within the gear pump cavity **206**, the gear pump **102** can drive the foam concentrate to be transferred from the gear pump inlet **124** to the gear pump outlet **205**.

(14) A Roots pump cavity **202** is formed in the Roots pump housing **116**, and a pair of rotors **203** are arranged in the Roots pump cavity **202**. The pair of rotors **203** have the same size and shape, and the cross-section of each rotor **203** is substantially in a shape of "8." A rotor shaft **213** is provided at the central position of each rotor **203**. Two rotor shafts **213** respectively extend along

the Z-axis direction and respectively constitute a rotation center of a corresponding one of the rotors **203**. The volume of the Roots pump cavity **202** is matched with the size of the pair of rotors **203**, so that the pair of rotors **203** can respectively rotate in the Roots pump cavity **202** around the corresponding rotor shafts **213** thereof. When the fire-fighting water with a certain flow rate flows from the Roots pump inlet **103** to the pair of rotors **203**, the pair of rotors **203** can be driven to rotate by the kinetic energy of the fire-fighting water, thus driving the Roots pump **101** to operate normally. During the normal operation of the Roots pump **101**, the pair of rotors **203** have relatively fixed rotation positions. However, since the cross-section of the rotors **203** is substantially in a shape of “8” and there are no intermeshing teeth or keys arranged between the pair of rotors **203**, the pair of rotors **203** cannot be intermeshed and positioned with each other, and thus it is impossible to ensure that correct relative positions of the two rotors **203** are maintained at every moment during the rotation process.

(15) In order to ensure the normal operation of the Roots pump **101**, a pair of synchronous gears **201** are arranged coaxially above the pair of rotors **203** along the Z-axis direction. That is to say, one synchronous gear **201** is arranged above the rotor shaft **213** of each rotor **203**, so that a corresponding one of rotors **203** and a corresponding one synchronous gear **201** can rotate synchronously. As shown in FIGS. **2**A and **2**B, the pair of synchronous gears **201** have the same size and shape, and are meshingly arranged at the same height. The above arrangement enables each rotor **203** to drive a corresponding one of synchronous gears **201** to rotate synchronously by means of the corresponding rotor shaft **213** thereof, and at the same time, the meshing rotation of the pair of synchronous gears **201** also affects the rotation position of the pair of rotors **203**, ensuring that the correct rotation position of the pair of rotors **203** is always maintained during the rotation process. In the embodiments of the present disclosure, the outer circumference of each synchronous gear **201** is provided with a plurality of fine gear teeth, and the provision of the fine gear teeth can ensure a stable meshing rotation of the pair of synchronous gears **201**, thus providing effective position guidance for the pair of rotors **201** during the operation of the Roots pump **101**.

(16) FIG. **3** is a transverse sectional view of the gear pump **102** in FIG. **1** at the positions of the gear pump inlet **124** and the gear pump outlet **205**, showing the structure of the gear pump **102** on a plane defined by the X-axis and the Y-axis. As shown in FIG. **3**, the cross-section of the gear pump cavity **206** is jointly defined by two mutually parallel straight side edges and two oppositely arranged semicircular arc side edges. Here, the two parallel straight side edges and two semicircular arc side edges respectively correspond to side walls of the gear pump cavity **206**. The two mutually parallel straight side edges comprise a left side edge **304** and a right side edge **305**, and the left side edge **304** and the right side edge **305** respectively extend along the Y-axis direction. The two semicircular arc side edges are respectively located on upper and lower sides of the Y-axis direction, comprising an upper side edge **306** and a lower side edge **307**, wherein the upper side edge **306** and the lower side edge **307** are respectively arranged to protrude outward. As shown in FIG. **3**, the gear pump inlet **124** is located in a middle position of the left side edge **304**, and the gear pump outlet **205** is located in a middle position of the right side edge **305**. Here, the gear pump inlet **124** penetrates through the side wall of the gear pump housing **113** corresponding to the left side edge **304**, and the gear pump outlet **205** penetrates through the side wall of the gear pump housing **113** corresponding to the right side edge **305**, so that the gear pump inlet **124** and the gear pump outlet **205** are respectively in fluid communication with the gear pump cavity **206**.

(17) The pair of gears **207** have the same size and shape, and are arranged side by side and top and bottom in the Y-axis direction. In the present disclosure, the gear **207** arranged above along the Y axis is defined as an upper gear **311**, and the gear **207** arranged below along the Y axis is defined as a lower gear **312**. The gear **207** of the present disclosure is a circular gear, and the shape of the gear **207** is matched with the shape of the gear pump cavity **206**. As shown in FIG. **3**, the arc diameters of the upper side edge **306** and the lower side edge **307** are respectively approximately the same as the diameters of the circles enclosed by the tips of the outer teeth of the gears **207**, so that the pair

of gears **207** can be accommodated in the gear pump cavity **206** and can meshingly rotate around their respective gear shafts **217**. When the foam proportional mixing device **100** is in a working state, the pair of gears **207** in the gear pump **102** rotate in a direction of an arrow shown in FIG. **3**. As shown in FIG. **3**, the pair of gears **207** rotate in opposite directions from each other. With the meshing rotation of the pair of gears **207**, when the gear teeth **303** of the upper gear **311** rotate to an upper half part, the tips of the gear teeth **303** are fitted with the corresponding side wall of the upper side edge **306**; and when the gear teeth **303** of the lower gear **312** rotate to a lower half part, the tips of the gear teeth **303** are fitted with the corresponding side wall of the lower side edge **307**. The structural arrangement between the pair of gears **207** and the side walls of the gear pump cavity **206** is such that: when the gear pump **102** is in a working state, the left outer circumference of meshing gears **207** and the side wall corresponding to the left side edge **304** form a left sealing area **301**, and the right outer circumference of meshing gears **207** and the right side wall of the gear pump cavity **206** form a right sealing area **302**.

(18) When the gear pump **102** is in a working state, at a position where the pair of gears **207** are meshed with each other, meshing gear teeth **303** located on the left side of the gear pump **102** are gradually disengaged from meshing, and gradually withdraw from the space between the teeth, so that the volume of the left sealing area **301** increases and a partial vacuum is formed. Since the gear pump inlet **124** is connected to the external foam concentrate tank, when the pressure of the left sealing area **301** of the gear pump **102** decreases, the foam concentrate in the foam concentrate tank will be driven by the pressure to enter the left sealing area **301** by means of the gear pump inlet **124** along the direction of the arrow shown in FIG. **3**. At this time, the meshing gear teeth **303** on the right side of the gear pump **102** gradually enter into meshing, so that the volume of the right sealing area **302** decreases. As the volume of the right sealing area **302** decreases, the foam concentrate in the right sealing area **302** is gradually squeezed out and discharged from the gear pump outlet **205** along the direction of the arrow in FIG. **3**. When the gear pump **102** rotates continuously, the gear teeth of the gear **207** of the left sealing area **301** are gradually disengaged from meshing, so that the left sealing area **301** continuously sucks the foam concentrate from the foam concentrate tank due to the increase of the sealing volume and the decrease of the pressure, and at the same time, the gear teeth of the gear **207** of the right sealing area **302** are gradually meshed, so that the right sealing area **302** continuously discharges the foam concentrate from the gear pump outlet **205** due to the decrease of the sealing volume.

(19) FIG. **4** is a transverse sectional view of the Roots pump **101** in FIG. **1** at the positions of the Roots pump inlet **103** and the Roots pump outlet **104**, showing the structure of the Roots pump **101** on the plane defined by the X-axis and the Y-axis. As shown in FIG. **4**, the cross-section of the Roots pump cavity **202** is also jointly defined by two mutually parallel straight side edges and two oppositely arranged semicircular arc side edges. Here, the two parallel straight side edges and the two semicircular arc side edges respectively correspond to the side walls of the Roots pump cavity **202**. The two mutually parallel straight side edges respectively extend along the Y-axis direction, and the two semicircular arc side edges respectively protrude outwards and are located on the upper and lower sides of the Y-axis direction. The Roots pump inlet **103** formed in the Roots pump inlet pipe **146** and the Roots pump outlet **104** formed in the Roots pump outlet pipe **147** respectively penetrate through the side walls of the Roots pump housing **116** corresponding to two parallel straight lines, so that the Roots pump inlet **103** and the Roots pump outlet **104** are respectively in fluid communication with the Roots pump cavity **202**. As shown in FIG. **4**, the Roots pump inlet **103** is located on the left side of the Roots pump **101**, and the Roots pump outlet **104** is located on the right side of the Roots pump **101**. An inlet channel **404** is formed at a position where the Roots pump inlet **103** interfaces with the Roots pump cavity **202**, and an outlet channel **405** is formed at a position where the Roots pump outlet **104** interfaces with the Roots pump cavity **202**. Here, the inlet channel **404** belongs to a part of the Roots pump inlet **103**, the outlet channel **405** belongs to a part of the Roots pump outlet **104**, and the inlet channel **404** and the outlet channel **405**

respectively face the middle position of the Roots pump cavity **202**.
(20) As shown in FIG. **4**, a pair of rotors **203** are adjacently arranged top and bottom in the Y-axis direction, and their corresponding two rotor shafts **213** are respectively arranged on the symmetry axis of the Roots pump cavity **202** extending in the Y-axis direction. The pair of rotors **203** comprise an upper rotor **421** and a lower rotor **422**, wherein the upper rotor **421** is located above along the Y-axis direction, and the lower rotor **422** is located below along the Y-axis direction. The connecting line between the top end and the bottom end of the rotor **203** of which the cross-section is in the shape of “8” is defined as the maximum penetration line D. The diameters of the two semicircular arcs defining the Roots pump cavity **202** are slightly greater than the length of the maximum penetration line D, so that the pair of rotors **203** can be accommodated in the Roots pump cavity **202** and can rotate around their respective rotor shafts **213**. In the state shown in FIG. **4**, two maximum penetration lines D of the pair of rotors **203** are perpendicular to each other, the maximum penetration line D of the upper rotor **421** extends along the Y-axis direction, the maximum penetration line D of the lower rotor **422** extends along the X-axis direction, and the position of the maximum penetration line D of the lower rotor **422** substantially coincides with the diameter position of semicircular arc on a lower side. Here, in the Y-axis direction, a lower end of the upper rotor **421** is just accommodated at the waist position where the lower rotor **422** is recessed inward.
(21) When the Roots pump **101** is in the working state, the fire-fighting water with a certain flow rate enters the Roots pump cavity **202** from the Roots pump inlet **103**, and drives the pair of rotors **203** to rotate along a direction of an arrow shown in FIG. **4**. As shown in FIG. **4**, under normal operation states, the pair of rotors **203** rotate in opposite directions relative to each other. The inventors of the present disclosure found that when the fire-fighting water with a certain flow rate directly enters the Roots pump cavity **202** by means of the Roots pump inlet **103**, part of the fire-fighting water can flow to the position in the Roots pump cavity **202** near side walls, so that the pair of rotors **203** are driven to rotate in opposite directions relative to each other. However, at this time, another part of the fire-fighting water will flow to the middle position of the Roots pump cavity **202**, thereby preventing the pair of rotors **203** from rotating in the direction of the arrow shown in FIG. **4**. That is to say, the inventors of the present disclosure found that when the rotors **203** are driven to rotate only by the kinetic energy of the fire-fighting water instead of being driven by a motor, the flow direction of the fire-fighting water in the Roots pump cavity **202** plays an important role in the normal rotation of the pair of rotors **203**. In order to ensure the normal rotation of the pair of rotors **203** in the Roots pump cavity **202**, the inventors of the present disclosure provide a fluid guide member **401** in the Roots pump inlet **103** formed in the Roots pump inlet pipe **146**, and the fire-fighting water is guided by the fluid guide member **401** to flow to the position of the inner wall of the Roots pump cavity **202**, so as to drive the pair of rotors **203** to rotate normally in the direction of the arrow shown in FIG. **4**.
(22) FIG. **5**A and FIG. **5**B respectively show the internal structure of the Roots pump lower housing **112** in FIG. **1** at different angles, for illustrating the structure of the fluid guide member **401**. As shown in FIG. **5**A and FIG. **5**B, the fluid guide member **401** extends in the Z-axis direction within the Roots pump inlet **103**. In this embodiment, the fluid guide member **401** is substantially in a triangular prism shape. The triangular prism-shaped guide member **401** comprises a top surface **501**, a bottom surface **502**, and three side surfaces **402**. As shown in FIG. **5**B, the top surface **501** is connected to a top wall of the Roots pump inlet **103**, and the bottom surface **502** is connected to a bottom wall of the Roots pump inlet **103**. One side surface **402** among the three side surfaces **402** is arranged on the inlet channel **404** of the Roots pump inlet **103**. It can be seen in conjunction with FIG. **4** and FIG. **5**B that the side surface **402** on the inlet channel **404** is approximately located in the middle position of the inlet channel **404** and is approximately flush with the side wall of the Roots pump cavity **202**. The other two side surfaces **402** are arranged substantially in the direction in which the fire-fighting water enters the Roots pump inlet **103**, forming a pair of fluid guide

surfaces **411** for guiding the flow direction of fire-fighting water. The pair of fluid guide surfaces **411** form a flow dividing edge **403** at a position where the pair of fluid guide surfaces **411** are connected to each other, and the flow dividing edge **403** extends between the top surface **501** and the bottom surface **502** along the Z-axis direction. As shown in FIG. **5**A, the flow dividing edge **403** faces the direction in which the fire-fighting water enters the Roots pump inlet **103**, and is arranged away from the Roots pump cavity **202**. The fluid guide surfaces **411** of the fluid guide member **401** are configured such that when the fluid flows from the Roots pump inlet **103** to the Roots pump cavity **202**, the pair of fluid guide surfaces **411** can guide the fluid to form two sub-flows, and the two sub-flows flow away from each other, so as to respectively drive the pair of rotors **203** to rotate in opposite directions from each other.
(23) As shown in FIG. **5**B, in order to avoid the fluid guide member **401**, a foam receiving port **117** is arranged at a distal end of the Roots pump inlet pipe **146**, approximately at a position outside the flow dividing edge **403**. The above arrangement enables the foam concentrate entering the Roots pump inlet **103** from the foam receiving port **117** to flow to the pair of fluid guide surfaces **411** of the fluid guide member **401** together with the fire-fighting water, and to flow together, under the guidance of the pair of fluid guide surfaces **411**, in the direction in which the pair of rotors **203** can be driven to work normally in the Roots pump cavity **202**. In the embodiment of the present disclosure, the foam receiving port **117** is arranged on the Roots pump inlet pipe **146** instead of on the side wall of the Roots pump cavity **202**. The above arrangement enables the foam concentrate to flow into the Roots pump cavity **202** together with the fire-fighting water, thus not interfering with the normal rotation of the pair of rotors **203** in the Roots pump cavity **202**. If the foam receiving port **117** is arranged on the side wall of the Roots pump cavity **202**, the foam concentrate from the foam receiving port **117** will directly flow into the Roots pump cavity **202**, where the flow direction of the foam concentrate is likely to be inconsistent with the rotation direction of the rotors **203** adjacent to the foam concentrate, thereby interfering with the normal rotation of the pair of rotors **203**.
(24) As shown in FIGS. **5**A and **5**B, the side surface **402** of the fluid guide member **401** that is arranged on the inlet channel **404** is opposite to the flow dividing edge **403**. The area of the side surface **402** opposite to the flow dividing edge **403** is defined as A, and the opening area of the inlet channel **404** of the Roots pump inlet **103** at the position of the inner wall of the Roots pump housing **116** is defined as S. In order to improve the flow guide effect of the fluid guide member **401**, not only to ensure the smooth flow of the fluid in the Roots pump inlet **103**, but also to effectively guide the flow direction of the fluid into the Roots pump cavity **202**, the area A of the side surface and the opening area S may satisfy: $\frac{1}{4} \leq A:S \leq \frac{3}{4}$.
(25) It can be seen in conjunction with FIG. **4** and FIGS. **5**A and **5**B that the cross-section of the fluid guide member **401** is an isosceles triangle. Here, the two legs of the isosceles triangle correspond to a pair of fluid guide surfaces **411**, the vertex of the isosceles triangle corresponds to the flow dividing edge **403**, and the base of the isosceles triangle corresponds to the side surface **402** opposite to the flow dividing edge **403**. As shown in FIG. **4**, the base of the isosceles triangle is located in the middle position of the inlet channel **404**. The length of the base of the isosceles triangle is defined as b, and the height of the isosceles triangle is defined as h. In order to effectively guide the fluid to flow toward the side wall of the Roots pump cavity **202** and ensure the flow guide effect of the fluid guide member **401**, the ratio h:b between the height h and the base b may satisfy: $\frac{1}{4} \leq h:b \leq 1$. In some embodiments, the ratio h:b between the height h and the base b may also satisfy: $\frac{1}{3} \leq h:b \leq \frac{1}{2}$. In the embodiments of the present disclosure, the fluid guide member **401** is in a triangular prism shape. In other embodiments, fluid guide members **401** with other shapes can also be provided, as long as the flow direction of the fluid in the Roots pump inlet **103** can be guided, and the effective driving of the fluid on the pair of rotors **203** in the Roots pump **101** can be realized.
(26) FIG. **6**A to FIG. **6**E respectively show the working states of the pair of rotors **203** in FIG. **4** in

one operation cycle, and describe the operation states of the Roots pump **101** from four typical stages. In the present disclosure, the position of the pair of rotors **203** shown in FIG. **4** is defined as an initial position, and the position of the pair of rotors **203** shown in FIG. **6**A is exactly the same as that in FIG. **4**. As shown in FIG. **6**A, when a pressure fluid flows into the Roots pump inlet **103** from the left side along a direction of an arrow, the fluid flows to a left area A jointly enclosed and formed by the pair of rotors **203** and the side wall of the Roots pump cavity **202**. Before the fluid enters the Roots pump cavity **202**, the fluid guide member **401** in the Roots pump inlet **103** can guide the fluid to respectively form two sub-flows flowing upward and downward, wherein the sub-flow flowing upward flows to the upper rotor **421** and the sub-flow flowing downward flows to the lower rotor **422**. The sub-flows flowing upward and downward respectively flow towards the side wall of the Roots pump cavity **202**, providing a driving force for the pair of rotors **203** to rotate in opposite directions.

(27) Under the rotation drive of the pressure fluid, the pair of rotors **203** rotate respectively in directions of arrows shown in FIG. **6**A, so as to rotate from the position in FIG. **6**A to the position in FIG. **6**B. During the rotation of the pair of rotors **203** from the position shown in FIG. **6**A to the position shown in FIG. **6**B, the maximum penetration line D of the upper rotor **421** rotates clockwise around the rotor shaft **213** thereof from a position extending in the Y-axis direction to a position inclined to the right, and the maximum penetration line D of the lower rotor **422** rotates counterclockwise from a position extending in the X-axis direction to a position inclined to the right. In the position shown in FIG. **6**B, the maximum penetration line D of the upper rotor **421** and the maximum penetration line D of the lower rotor **422** are parallel to each other. The lower right side of the upper rotor **421** is in contact with the upper left side of the lower rotor **422**, and the upper rotor **421**, lower rotor **422**, and the left side of the side wall of the Roots pump cavity **202** are jointly enclosed to form a left area B. With the rotation of the upper rotor **421** and the lower rotor **422**, the fluid flowing from the Roots pump inlet **103** into the left area A in FIG. **6**A gradually moves to the left area B in FIG. **6**B.

(28) As the pressure fluid continuously flows from the Roots pump inlet **103** into the Roots pump cavity **202**, the pair of rotors **203** continuously obtain the rotational kinetic energy from the pressure fluid, and then rotate from the position in FIG. **6**B to the positions in FIG. **6**C, FIG. **6**D, and FIG. **6**E in sequence. In the rotation process from the position in FIG. **6**B to the position in FIG. **6**C, the maximum penetration line D of the upper rotor **421** rotates clockwise from a position inclined to the right to a position extending in the X-axis direction, and the maximum penetration line D of the lower rotor **422** rotates counterclockwise from a position inclined to the right to a position extending in the Y-axis direction. In the position shown in FIG. **6**C, the upper rotor **421** and the side wall of the Roots pump cavity **202** are jointly enclosed to form an upper area C. With the rotation of the upper rotor **421** and the lower rotor **422**, the fluid in the left area B in FIG. **6**B gradually moves to the upper area C in FIG. **6**C.

(29) In the position shown in FIG. **6**D, the maximum penetration line D of the upper rotor **421** and the maximum penetration line D of the lower rotor **422** are parallel to each other, and are inclined to the left respectively. The lower left side of the upper rotor **421** is in contact with the upper right side of the lower rotor **422**, and the upper rotor **421**, the lower rotor **422**, and the right side of the side wall of the Roots pump cavity **202** are jointly enclosed to form a right area E. With the rotation of the upper rotor **421** and the lower rotor **422**, the fluid in the upper area C in FIG. **6**C gradually moves to the right area E in FIG. **6**D. As shown in FIG. **6**D, the right area E is in communication with the Roots pump outlet **104**.

(30) During the rotation process of the pair of rotors **203** from the position shown in FIG. **6**D to the position shown in FIG. **6**E, the maximum penetration line D of the upper rotor **421** rotates from a position inclined to the left to a position extending in the Y-axis direction, and the lower rotor **422** rotates from a position inclined to the left to a position extending in the X-axis direction. In the position shown in FIG. **6**E, the upper rotor **421**, the lower rotor **422**, and the right side of the side

wall of the Roots pump cavity **202** are jointly enclosed to form a right area F. With the rotation of the upper rotor **421** and the lower rotor **422**, the fluid in the right area E in FIG. **6**D gradually moves to the right area F in FIG. **6**E. Since the right area E and the right area F are respectively in communication with the Roots pump outlet **104**, and the volume of the right area F is less than that of the right area E, the fluid is continuously discharged from the Roots pump outlet **104** due to being squeezed during the movement process from the right area E to the right area F.
(31) In one rotation cycle shown in FIG. **6**A to FIG. **6**E, the fluid flowing from the Roots pump inlet **103** into the Roots pump cavity **202** gradually flows to the Roots pump outlet **104** along with the rotation of the pair of rotors **203**. With reference to FIG. **6**A to FIG. **6**E, it can be seen that the position of the pair of rotors **203** in FIG. **6**E is exactly the same as that in FIG. **6**A, and the pair of rotors **203** in FIG. **6**A can return to the initial position shown in FIG. **6**A by means of a series of rotation movements in FIG. **6**A to FIG. **6**E, so as to continue the rotation movement in the next cycle.
(32) FIG. **7**A and FIG. **7**B are respectively longitudinal sectional views of the foam proportional mixing device **100** in FIG. **1** at different angles. Here, FIG. **7**A shows a sectional view of the foam proportional mixing device **100** in the plane defined by the X-axis and the Z-axis, and FIG. **7**B shows a sectional view of the foam proportional mixing device **100** in the plane defined by the Y-axis and the Z-axis. As shown in FIG. **7**A and FIG. **7**B, a pair of gears **207** are located at the same height of the Z-axis, a pair of synchronous gears **201** are located at the same height of the Z-axis, and a pair of rotors **203** are located at the same height of the Z-axis, which are respectively arranged in the Z-axis direction from top to bottom. In order to ensure that the pair of rotors **203** maintain the correct positions at each moment during the rotation process, one synchronous gear **201** is coaxially arranged with one rotor **203** and the other synchronous gear **201** is coaxially arranged with the other rotor **203** among two synchronous gears **201** and two rotors **203**. In order to ensure the proportional mixing of fire-fighting water and foam concentrate, the rotor shaft **213** of one rotor **203** of the Roots pump **101** is connected with the gear shaft **217** of one gear **207** of the gear pump **102** in the foam proportional mixing device **100**, so as to drive the gear **207** in the gear pump **102** to rotate by means of one rotor **203** in the Roots pump **101**. In order to realize the coaxial connection between the rotor shaft **213** and the gear shaft **217**, a coupling **703** is provided in the embodiments of the present disclosure between one of the rotor shafts **213** and one gear shaft **217** corresponding thereto. As shown in FIG. **7**B, the coupling **703** is located above the synchronous gear **201** and connected between the rotor shaft **213** of the upper rotor **421** and the gear shaft **217** of the upper gear **311**. However, the rotor shaft **213** of the lower rotor **422** and the gear shaft **217** of the upper gear **312** are staggered from each other without connection. The coaxial connection between the rotor shaft **213** and the gear shaft **217** enables the Roots pump **101** and the gear pump **102** to rotate synchronously.
(33) The operation steps of the foam proportional mixing device **100** are as follows: when the foam proportional mixing device **100** starts to operate, the fire-fighting water supply end supplies the fire-fighting water with a certain flow rate from the Roots pump inlet **103** to the Roots pump **101**. Under the guidance of the fluid guide member **401**, the fire-fighting water in the Roots pump inlet **103** forms a specific flow direction, thereby driving the pair of rotors **203** to respectively rotate in opposite directions around their respective rotor shafts **213**. During the rotation process of the pair of rotors **203**, the synchronous meshing rotation of the pair of synchronous gears **201** coaxially arranged with the pair of rotors **203** causes the pair of rotors **203** to be always kept in the correct rotation position. With the rotation of the pair of rotors **203**, the upper gear **311** coaxially connected with the upper rotor **421** also rotates therewith. By means of the meshing relationship between the upper gear **311** and the lower gear **312**, the rotation of the upper gear **311** can drive the lower gear **312** to rotate synchronously in an opposite direction. The gear pump inlet **124** is in communication with the foam concentrate tank, and therefore, with the reverse rotation of the pair of gears **207**, the gear pump **102** can draw the foam concentrate from the foam concentrate tank by means of the gear

pump inlet **124**, and deliver the drawn foam concentrate to the foam receiving port **117** on the Roots pump inlet pipe **146** by means of the gear pump outlet **205**. After entering the Roots pump inlet **103** from the foam receiving port **117**, the foam concentrate flows jointly with the fire-fighting water toward the Roots pump cavity **202** under the driving of the flow velocity of the fire-fighting water. Similarly, the fire-fighting water mixed with the foam concentrate will also be subjected to the guide function of the fluid guide member **401** during the process of the fire-fighting water flowing from the Roots pump inlet **103** towards the Roots pump cavity **202**. The fire-fighting water mixed with the foam concentrate can further drive the pair of rotors **203** to rotate continuously after being guided. With the rotation of the pair of rotors **203**, the fire-fighting water mixed with the foam concentrate flows into the Roots pump cavity **202** and is fully mixed in the Roots pump cavity **202** to form a foam solution. The formed foam solution is gradually discharged from the Roots pump outlet **104** with the rotation of the pair of rotors **203**. In the embodiments of the present disclosure, the Roots pump outlet **104** is in communication with an external fire-fighting pipe. Since the gear pump **102** and the Roots pump **101** always rotate synchronously, the foam proportional mixing device **100** of the present disclosure can always mix the fire-fighting water and the foam concentrate at a stable ratio.

(34) In one aspect, in the present disclosure, the Roots pump **101** is applied to the foam proportional mixing device **100**, and by utilizing the advantage of the small size of the Roots pump **101**, the foam proportional mixing device **100** prepared with the Roots pump **101** has a simple and compact structure. In another aspect, in the present disclosure, a fluid guide member **401** having a specific structure is provided in the Roots pump inlet **103** of the Roots pump **101**, and the flow direction of the fluid flowing into the Roots pump cavity **202** is controlled by means of the fluid guide member **401**, thereby utilizing the pressure of the fluid itself to drive the pair of rotors **203** in the Roots pump **101** to rotate effectively. The foam proportional mixing device **100** of the present disclosure can realize the stable mixing and delivery of the fire-fighting water and the foam concentrate according to a certain ratio only by means of the action of the pressure fluid, without the need for additional power, and has the advantages of large outlet flow, small pressure loss, and convenient and quick use. In addition, due to its small and compact size, the foam proportional mixing device **100** of the present disclosure can be installed in vertical and horizontal pipes, so as to be suitable for fire-fighting work on various occasions.

(35) Although the present disclosure has been described in conjunction with the examples of embodiments outlined above, various alternatives, modifications, variations, improvements, and/or substantial equivalents, whether known or foreseeable now or in the near future, may be obvious to those having at least ordinary skill in the art. Accordingly, the examples of embodiments of the present disclosure, as set forth above, are intended to be illustrative, not limiting. Various changes may be made without departing from the spirit or scope of the present disclosure. Accordingly, the present disclosure is intended to embrace all known or earlier developed alternatives, modifications, variations, improvements, and/or substantial equivalents. The technical effects and technical problems in the present specification are exemplary and not limiting. It should be noted that the embodiments described in the present specification may have other technical effects and may solve other technical problems.

Claims

1. A foam proportional mixing device, comprising: a Roots pump, the Roots pump comprising: a Roots pump housing, wherein a Roots pump cavity is formed in the Roots pump housing; a Roots pump inlet pipe connected to an outer side of the Roots pump housing and in fluid communication with the Roots pump cavity; a foam receiving port arranged on the Roots pump inlet pipe; a pair of rotors, the pair of rotors being located in the Roots pump cavity; and a fluid guide member, wherein the fluid guide member is arranged in the Roots pump inlet pipe, and is configured to guide a fluid

to flow to the pair of rotors and to provide driving forces to the pair of rotors for rotating same in opposite directions from each other; and a gear pump, the gear pump comprising: a gear pump housing, wherein a gear pump cavity is formed in the gear pump housing, a gear pump inlet, and a gear pump outlet; and a pair of gears located within the gear pump cavity, one of the pair of gears connected to one of the pair of rotors via at least one shaft; wherein the foam receiving port is in communication with the gear pump outlet, such that the gear pump outlet is in fluid communication with the Roots pump cavity.
2. The foam proportional mixing device according to claim 1, wherein the Roots pump further comprises a Roots pump inlet at least partially formed in the Roots pump inlet pipe and a Roots pump outlet, wherein the Roots pump inlet and the Roots pump outlet are respectively arranged on two opposite sides of the Roots pump housing, and the Roots pump inlet and the Roots pump outlet are respectively in communication with the Roots pump cavity.
3. The foam proportional mixing device of claim 1, wherein the foam proportional mixing device is configured such that a rotary shaft of one rotor in the pair of rotors of the Roots pump is connected to a gear shaft of one gear in the pair of gears of the gear pump, such that the one rotor drives the one gear to rotate, the foam proportional mixing device further comprising: a coupling connected between the rotary shaft of the one rotor and the gear shaft of the one gear, and the coupling, the rotary shaft, and the gear shaft are arranged coaxially, so that the one rotor can drive the one gear to rotate via the coupling.
4. The foam proportional mixing device of claim 3, wherein: rotary shafts of the pair of Roots pump rotors both extend along a height of the Roots pump housing, and the fluid guide member extends along the height of the Roots pump housing.
5. The foam proportional mixing device of claim 2, wherein: the fluid guide member comprises a pair of fluid guide surfaces, wherein the pair of fluid guide surfaces are configured as follows: when a fluid flows from the Roots pump inlet to the Roots pump cavity, the pair of fluid guide surfaces guide the fluid to form two sub-flows, and the two sub-flows flow away from each other to respectively drive the pair of rotors to rotate in opposite directions from each other.
6. The foam proportional mixing device of claim 5, wherein: the fluid guide member is substantially in a triangular prism shape, and the fluid guide member comprises a top surface, a bottom surface, a flow dividing edge extending between the top surface and the bottom surface, and a pair of side surfaces connected to two opposite sides of the flow dividing edge, the pair of side surfaces forming the pair of fluid guide surfaces; wherein the top surface and the bottom surface are respectively connected to an inner wall of the Roots pump inlet, and the flow dividing edge is arranged away from the Roots pump cavity.
7. The foam proportional mixing device of claim 6, wherein: the fluid guide member further comprises a side surface opposite to the flow dividing edge, and the side surface opposite to the flow dividing edge is flush with an inner wall of the Roots pump housing at the Roots pump inlet; and an area of the side surface opposite to the flow dividing edge is A, an opening area of the Roots pump inlet at the inner wall of the Roots pump housing is S, and the area A of the side surface and the opening area S satisfy: $\frac{1}{4} \leq A:S \leq \frac{3}{4}$.
8. The foam proportional mixing device of claim 6, wherein: a cross-section of the fluid guide member is an isosceles triangle, the pair of fluid guide surfaces correspond to two legs of the isosceles triangle, a length of a base of the isosceles triangle is b, a height of the isosceles triangle is h, and a ratio h:b between the base b and the height h satisfies: $\frac{1}{3} \leq h:b \leq \frac{1}{2}$.
9. The foam proportional mixing device of claim 2, wherein: the gear pump inlet is configured to receive a foam concentrate, and the gear pump is configured to suck in the foam concentrate from the gear pump inlet and discharge the foam concentrate from the gear pump outlet; and the Roots pump inlet is configured to receive a pressure fluid and the foam concentrate from the gear pump outlet, and the Roots pump is configured to mix the pressure fluid and the foam concentrate flowing in from the Roots pump inlet pipe in the Roots pump cavity and discharge same from the

Roots pump outlet.
10. A foam proportional mixing device, wherein the foam proportional mixing device comprises: a pump comprising: a housing, wherein a cavity is formed in the housing; an inlet pipe connected to an outer side of the housing and in fluid communication with the cavity; a foam receiving port arranged on the inlet pipe; and a pair of rotors, the pair of rotors being located in the cavity; and a gear pump, the gear pump comprising: a gear pump housing, wherein a gear pump cavity is formed in the gear pump housing, a gear pump inlet, and a gear pump outlet; and a pair of gears located within the gear pump cavity, one of the pair of gears connected to one of the pair of rotors via at least one shaft; wherein the foam receiving port is in communication with the gear pump outlet, such that the gear pump outlet is in fluid communication with the cavity.
11. The foam proportional mixing device of claim 10, wherein the pump further comprises: an inlet and an outlet, wherein the inlet and the outlet are respectively arranged on two opposite sides of the housing, and the inlet and the outlet are respectively in communication with the cavity; and a fluid guide member arranged in the inlet and configured to (a) guide a fluid to flow to the pair of rotors and to (b) provide driving forces to the pair of rotors for rotating same in opposite directions from each other.
12. The foam proportional mixing device of claim 10 further comprising: a coupling connected between a rotary shaft of one rotor of the pair of rotors and a gear shaft of one gear of the pair of gears, and the coupling, the rotary shaft, and the gear shaft are arranged coaxially, so that the one rotor can drive the one gear to rotate via the coupling.
13. The foam proportional mixing device of claim 10, wherein the pair of rotors include a pair of rotary shafts that extend along a height of the housing.
14. The foam proportional mixing device of claim 11, wherein: the fluid guide member comprises a pair of fluid guide surfaces, wherein the pair of fluid guide surfaces are configured as follows: when a fluid flows from the inlet to the cavity, the pair of fluid guide surfaces guide the fluid to form two sub-flows, and the two sub-flows flow away from each other to respectively drive the pair of rotors to rotate in opposite directions from each other.
15. The foam proportional mixing device of claim 14, wherein: the fluid guide member is substantially in a triangular prism shape, and the fluid guide member comprises a top surface, a bottom surface, a flow dividing edge extending between the top surface and the bottom surface, and a pair of side surfaces connected to two opposite sides of the flow dividing edge, the pair of side surfaces forming the pair of fluid guide surfaces; wherein the top surface and the bottom surface are respectively connected to an inner wall of the inlet, and the flow dividing edge is arranged away from the cavity.
16. The foam proportional mixing device of claim 15, wherein: the fluid guide member further comprises a side surface opposite to the flow dividing edge, and the side surface opposite to the flow dividing edge is flush with an inner wall of the housing at the inlet; and an area of the side surface opposite to the flow dividing edge is A, an opening area of the inlet at the inner wall of the housing is S, and the area A of the side surface and the opening area S satisfy: $\frac{1}{4} \leq A:S \leq \frac{3}{4}$.
17. The foam proportional mixing device of claim 16, wherein: a cross-section of the fluid guide member is an isosceles triangle, the pair of fluid guide surfaces correspond to two legs of the isosceles triangle, a length of a base of the isosceles triangle is b, a height of the isosceles triangle is h, and a ratio h:b between the base b and the height h satisfies: $\frac{1}{3} \leq h:b \leq \frac{1}{2}$.
18. A Roots pump for a foam proportional mixing device, the Roots pump comprising: a Roots pump housing, wherein a Roots pump cavity is formed in the Roots pump housing; a Roots pump inlet and a Roots pump outlet, wherein the Roots pump inlet and the Roots pump outlet are respectively arranged on two opposite sides of the Roots pump housing, and the Roots pump inlet and the Roots pump outlet are respectively in communication with the Roots pump cavity; a Roots pump inlet pipe connected to an outer side of the Roots pump housing; a foam receiving port arranged on the Roots pump inlet pipe and in communication with an outlet of a gear pump, such

that the outlet of the gear pump is in fluid communication with the Roots pump cavity; a pair of rotors, the pair of rotors being located in the Roots pump cavity; and a fluid guide member, wherein the fluid guide member is arranged in the Roots pump inlet, and is configured to guide a fluid to flow to the pair of rotors and to provide driving forces to the pair of rotors for rotating same in opposite directions from each other.